

Chatbot Response Depth Control

```
import time
```

```
# Knowledge base (you can expand this)
```

```
knowledge_base = {
```

```
    "ai": {
```

```
        "short": "AI is the simulation of human intelligence by machines.",
```

```
        "medium": "Artificial Intelligence (AI) refers to machines that can perform tasks requiring  
human intelligence like learning, reasoning, and decision-making.",
```

```
        "deep": """Artificial Intelligence (AI) is a branch of computer science that focuses on  
creating systems capable of performing tasks that normally require human intelligence.
```

```
It includes:
```

- Machine Learning
- Natural Language Processing
- Computer Vision

```
Example: Chatbots, self-driving cars, recommendation systems."""
```

```
    },
```

```
    "python": {
```

```
        "short": "Python is a high-level programming language.",
```

```
        "medium": "Python is an easy-to-learn programming language used in web development,  
AI, data science, and automation.",
```

```
        "deep": """Python is a powerful, interpreted programming language known for its simplicity  
and readability.
```

```
Features:
```

- Easy syntax
- Large libraries (NumPy, TensorFlow)
- Cross-platform

```
Used in:
```

- AI & ML
- Web apps (Django, Flask)
- Automation scripts"""

```
    }
```

```
}
```

```
def get_response(user_input, depth):
```

```
    user_input = user_input.lower()
```

```

for key in knowledge_base:
    if key in user_input:
        if depth == 1:
            return knowledge_base[key]["short"]
        elif depth == 2:
            return knowledge_base[key]["medium"]
        elif depth == 3:
            return knowledge_base[key]["deep"]

return "Sorry, I don't understand that yet."

```

```

def chatbot():
    print("🤖 Chatbot with Response Depth Control")
    print("Type 'exit' to quit\n")

    while True:
        user_input = input("You: ")

        if user_input.lower() == "exit":
            print("Bot: Goodbye!")
            break

        try:
            depth = int(input("Choose response depth (1=short, 2=medium, 3=deep): "))
        except:
            print("Invalid input. Defaulting to medium.")
            depth = 2

        print("Bot is thinking...")
        time.sleep(1) # simulate real-time delay

        response = get_response(user_input, depth)
        print("Bot:", response)
        print("-" * 50)

# Run chatbot
chatbot()

```